Winning the Space Race with Data Science

SpaceX Falcon 9 First Stage Landing Prediction

IBM Applied Data Science Capstone · Eyad Alarfaj · July 2026

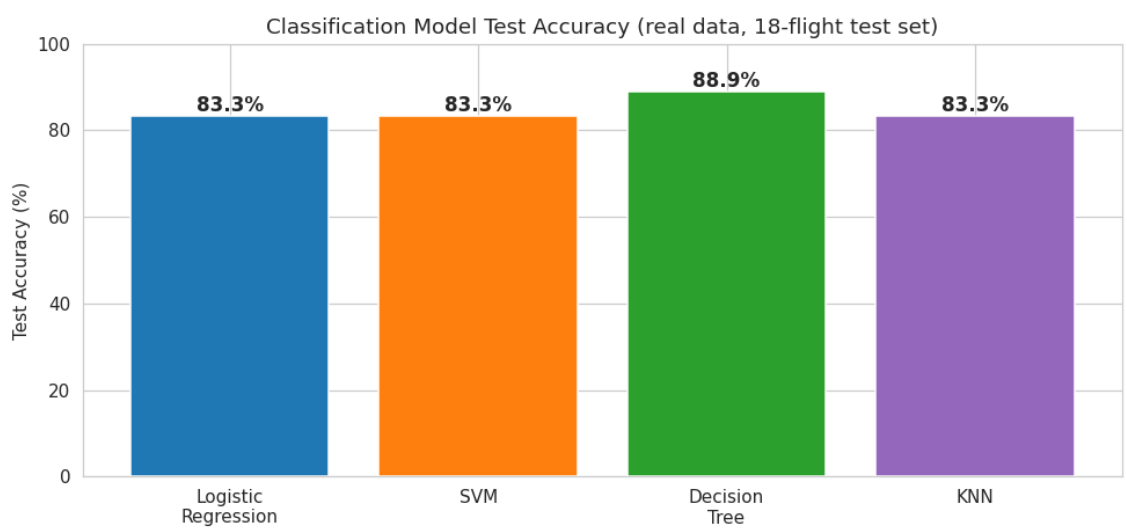
Built on the real IBM SpaceX dataset — 90 Falcon 9 launches, 2010–2020

GitHub: https://github.com/Eyad-Alarfaj/spacex-capstone

Outline — Assignment Sections Completed

Executive Summary

- **Methodologies:** collected 90 Falcon 9 launches from the SpaceX API and Wikipedia; wrangled the data into a landing-success target (Class); ran EDA with visualization and SQL; built a Folium map and a Plotly dashboard; and trained four classification models with GridSearchCV.
- **Results:** the first stage landed successfully on 66.7% of flights (60 of 90); success rose from ~0% in 2013 to ~90% by 2019–2020; the best predictor is the Decision Tree, reaching 88.89% accuracy on the held-out test set.
- **All notebooks, data, dashboard, and model files:** https://github.com/Eyad-Alarfaj/spacex-capstone



Introduction

- **Project background:** SpaceX advertises Falcon 9 launches at $62M versus $165M+ for competitors, largely because it recovers and re-flies the first stage. Whether the first stage lands therefore determines the true cost of a launch.
- **Problem:** predict whether the Falcon 9 first stage will land successfully, so a competitor can estimate cost and bid against SpaceX.
- **Questions:** (1) How do payload, orbit, launch site and booster reuse affect landing success? (2) Has success improved over time? (3) Which classification model predicts landing best?
- **Dataset:** 90 Falcon 9 launches, 2010-06-04 to 2020-11-05, across 3 launch sites and 11 orbit types — 60 landed, 30 failed.

Methodology — Data Collection

- **Sources:** SpaceX REST API (/v4/launches) for launch, booster, payload and core data, plus web-scraping of the Wikipedia Falcon 9 launch table to fill gaps.
- **Fields:** flight number, date, booster version, payload mass, orbit, launch site, landing outcome, number of flights, grid fins, reuse, legs, landing pad, block, reused count, serial, and coordinates.
- **Process:** request JSON → normalize into a pandas DataFrame → filter to Falcon 9 → clean payloads and outcomes → export CSV.
- **Result:** 90 clean launches. Notebooks: data-collection-api and web-scraping in the repository.

Methodology — Data Wrangling

- **Target creation:** the 8 distinct landing outcomes were mapped to a binary Class — 1 if the booster landed (True ASDS/RTLS/Ocean), 0 otherwise (None, False, or None ASDS).
- **Class balance:** 60 successes / 30 failures (66.7% success).
- **Feature prep:** computed success rate by orbit; one-hot encoded Orbit, LaunchSite, LandingPad and Serial for modeling; kept FlightNumber, PayloadMass, Flights, Block, ReusedCount, GridFins, Reused, Legs.
- **Notebook:** data-wrangling in the repository (executed).

Methodology — EDA with Data Visualization

- **Charts and rationale:** FlightNumber vs LaunchSite and vs Orbit scatter (does experience raise success?); PayloadMass vs Orbit scatter (which heavy-payload orbits land?); success-rate bar by orbit; and the yearly success-rate line (the learning curve).
- **Tools:** matplotlib and seaborn (eda-dataviz notebook). Every chart in this deck is computed from the 90-launch dataset.
- **Approach:** pair each operational variable with the landing outcome to surface the true drivers before modeling.

Methodology — EDA with SQL

- **Setup:** the dataset was loaded into a Db2 / SQLite table and queried with SQL magic (%sql) in the eda-sql notebook.
- Q1 unique launch sites; Q2 total payload mass; Q3 average payload per booster version; Q4 first successful ground-pad (RTLS) landing date; Q5 boosters with drone-ship (ASDS) success; Q6 counts of each landing outcome; Q7 success/failure by orbit.
- Results are shown on the EDA-with-SQL results slide (slide 17).

Methodology — Interactive Map with Folium

- **Objects:** a folium.Map plus a CircleMarker per launch site sized by launch count and colored by success rate, with popups reporting each site's launches and success percentage; markers use the real site coordinates from the dataset.
- **Questions answered:** where do launches concentrate, and which coastal sites perform best?
- **Output:** spacex_map.html (interactive). Map on slide 18.

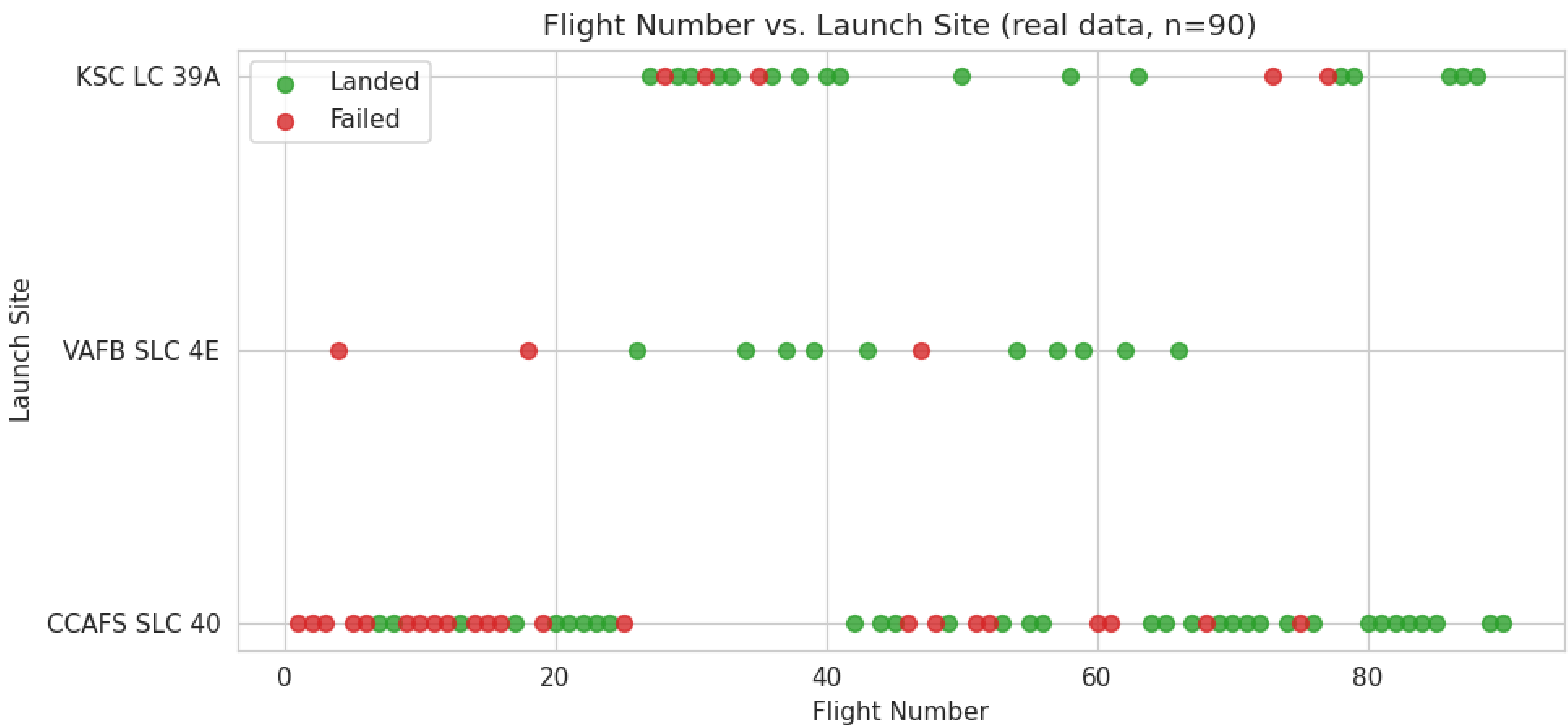
Methodology — Plotly Dashboard

- **Panels:** successful-landings share by site (pie), success rate by orbit (bar), yearly success trend (line), and success rate by site (bar).
- **Interactivity:** hover tooltips give exact values; the site and payload views mirror the course Plotly Dash app.
- **Output:** spacex_dash_dashboard.html. Dashboard on slide 19.

Methodology — Predictive Analysis (Classification)

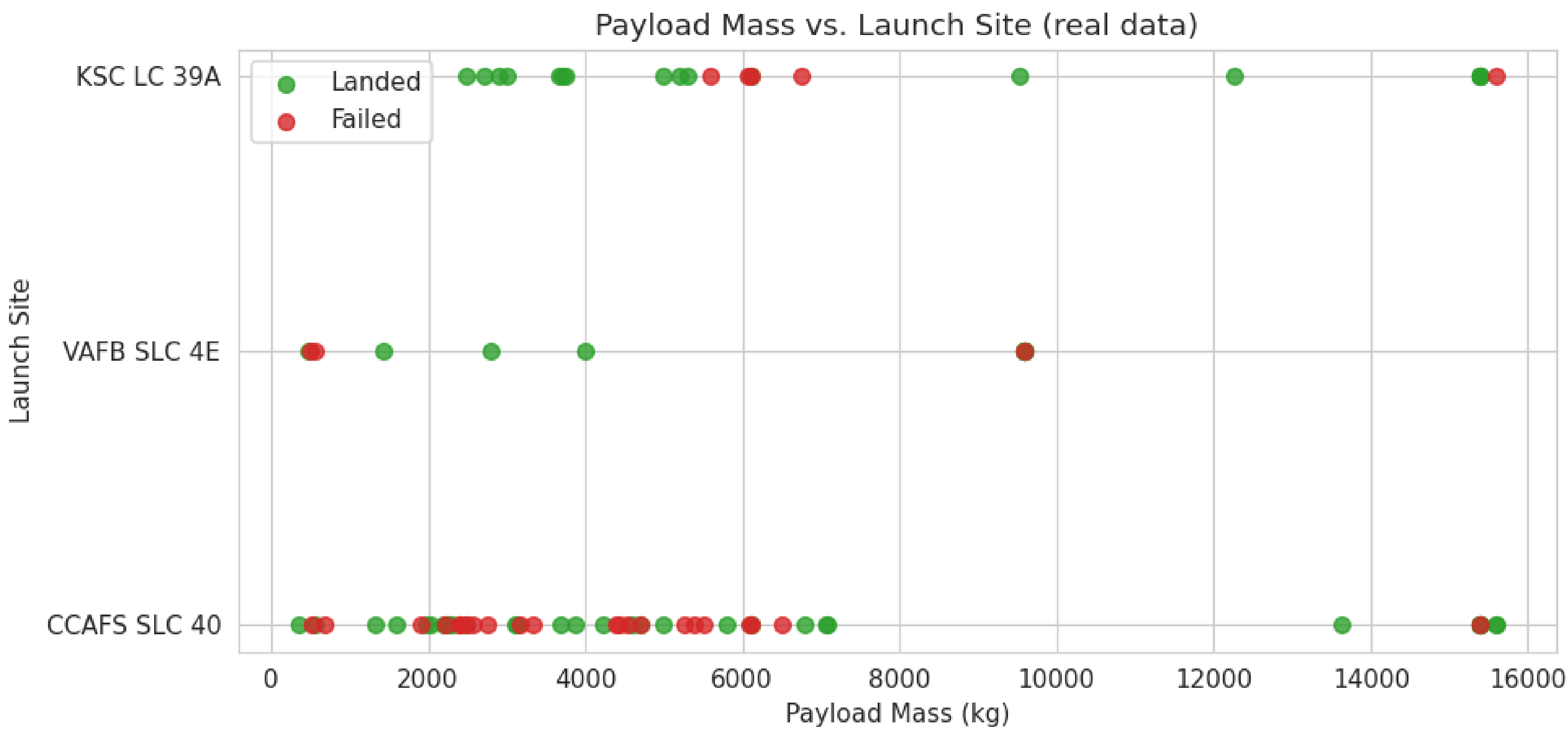
- **Pipeline:** standardize features with StandardScaler → 80/20 train-test split (72 train / 18 test) → tune each model with GridSearchCV (10-fold CV) → evaluate on the held-out test set.
- **Models:** Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbors.
- **Metrics:** cross-validation best score, test accuracy, and the confusion matrix.
- **Selection:** the model with the highest test accuracy wins — here the Decision Tree. Notebook: predictive-analysis (executed).

EDA Results — Flight Number vs. Launch Site



Failures dominate the early flights (low flight numbers); after ~flight 40 almost every launch lands, at all three sites.
CCAFS SLC 40 carries the most volume (55 of 90 launches) and shows the clearest early-to-late improvement.

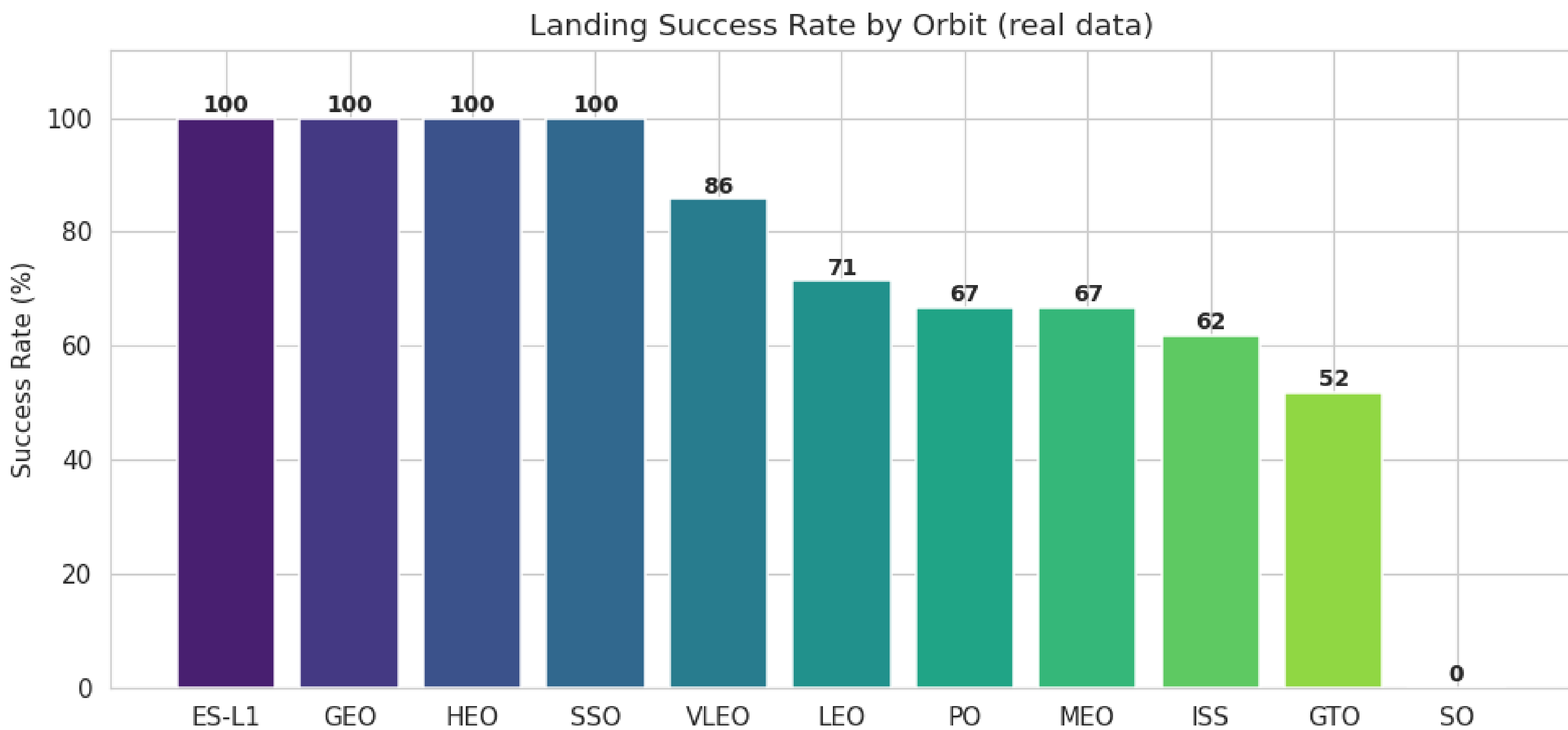
EDA Results — Payload Mass vs. Launch Site



Heavy payloads (>9,000 kg) are almost all successes — these are the later Starlink/VLEO missions flown on reused, flight-proven boosters.

Landed flights actually average heavier payloads (6,765 kg) than failures (4,784 kg): experience, not light payloads, drives success.

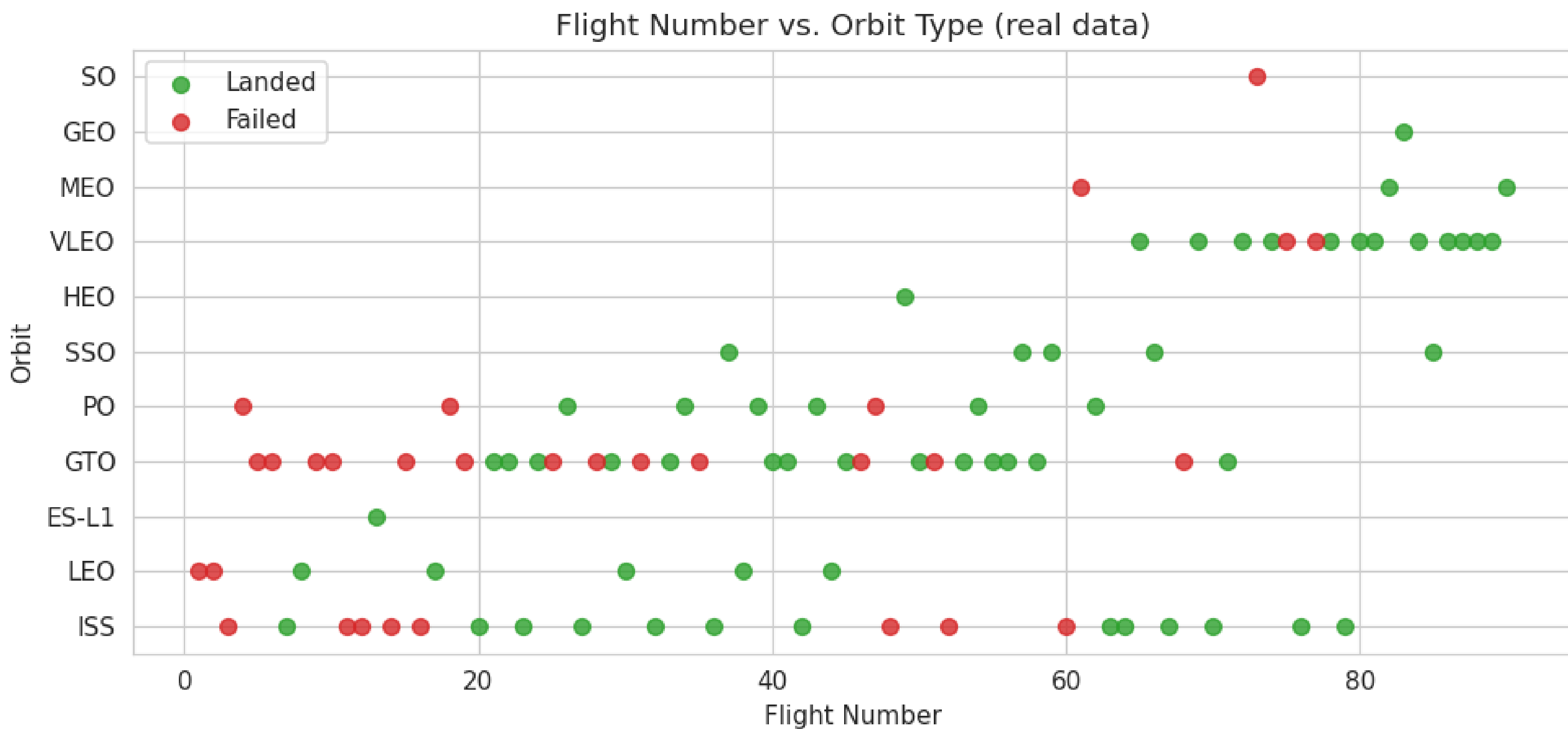
EDA Results — Success Rate by Orbit



ES-L1, GEO, HEO and SSO show 100% success (but small samples); VLEO is strong at 86% over 14 flights.

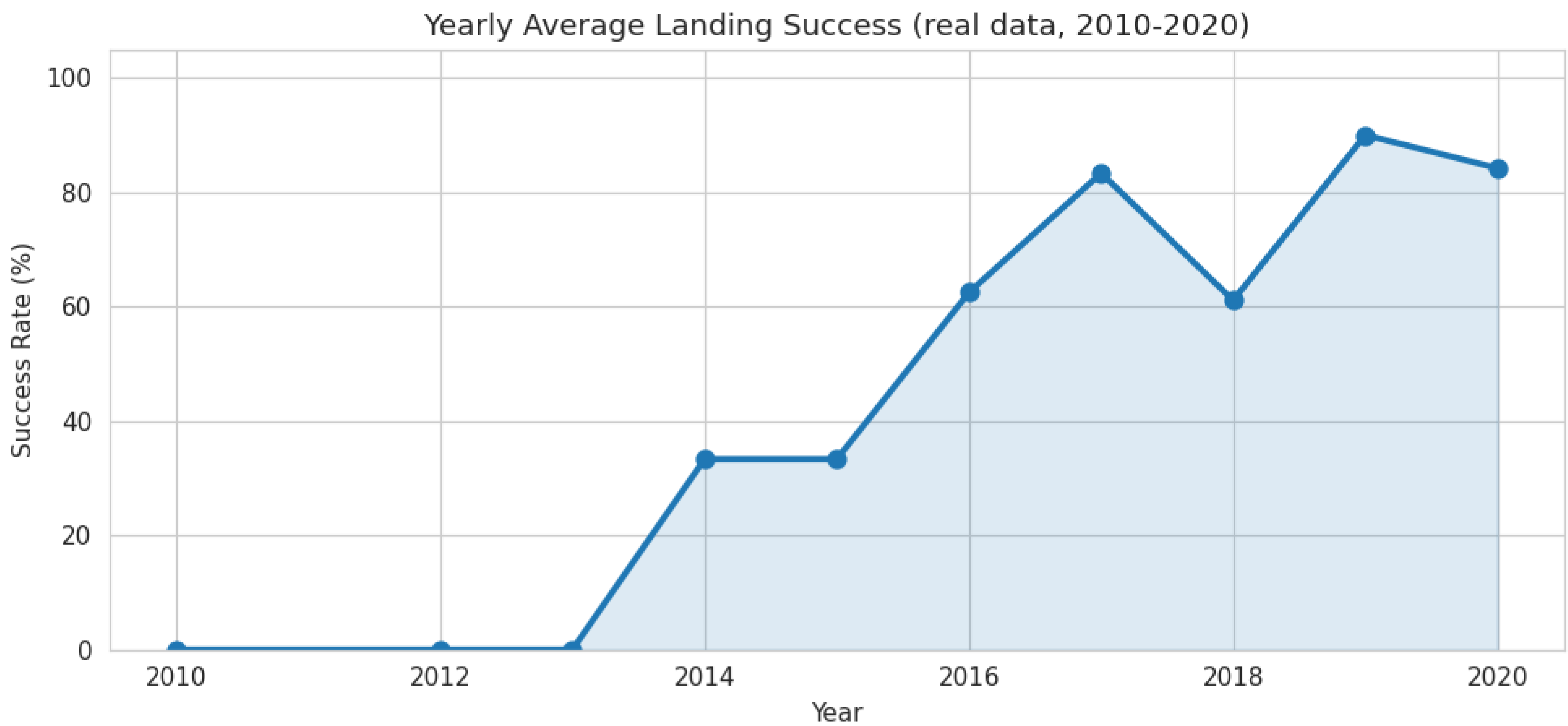
GTO is the hardest common orbit at 52% (27 flights) — high-energy transfers leave little fuel for landing; ISS sits at 62%.

EDA Results — Flight Number vs. Orbit



LEO and GTO failures cluster in the early program; the LEO/VLEO Starlink era (high flight numbers) is almost all green.
Newer orbits (VLEO) appear only late and land reliably, reflecting mature operations.

EDA Results — Yearly Success Trend



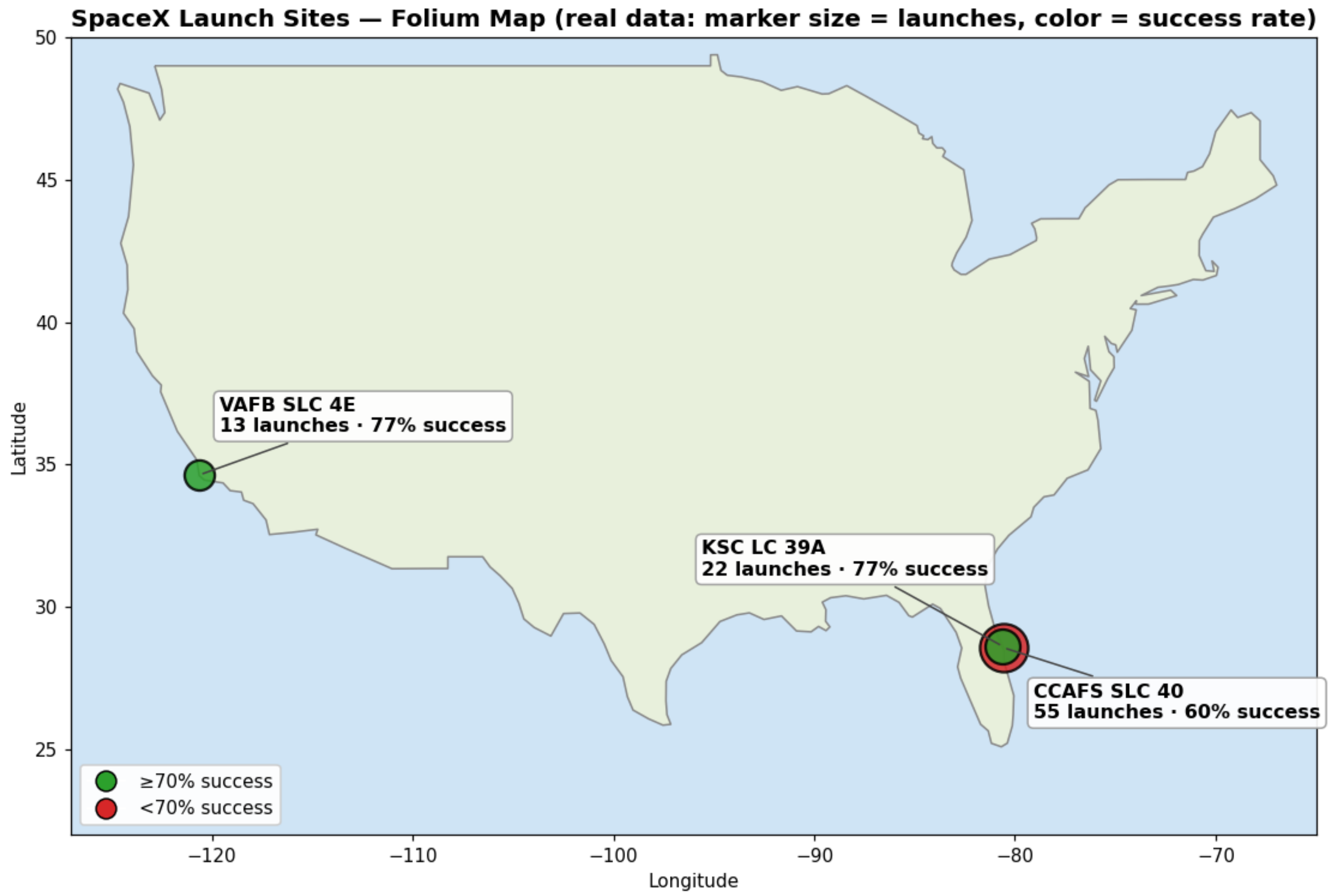
Success climbs from 0% (2010–2013) through 33% (2014–15) to 90% in 2019 and 84% in 2020 — a clear learning curve.
This is why FlightNumber is such a strong feature: it proxies accumulated engineering experience.

EDA with SQL — Results

SQL Query	Result
1. Unique launch sites	CCAFS SLC 40, KSC LC 39A, VAFB SLC 4E
2. Total payload mass	549,446 kg across 90 launches
3. Avg payload by site	CCAFS 5548 · KSC 7606 · VAFB 5919 kg
4. First ground-pad (RTLS) landing	2015-12-22 (Flight 17, CCAFS SLC 40)
5. Successful drone-ship (ASDS) landings	41 flights
6. Success rate by site	KSC 77% · VAFB 77% · CCAFS 60%
7. Landing outcomes	60 landed / 30 failed of 90

SQL confirms the visual EDA: KSC and VAFB outperform CCAFS, drone-ship recovery (41 successes) is the workhorse method, and the first ground-pad landing (Dec 2015) marks the start of reliable recovery.

Interactive Map with Folium — Results



All three sites sit on southern coasts by open ocean. KSC LC 39A (77%) and VAFB (77%) outperform the high-volume CCAFS (60%).

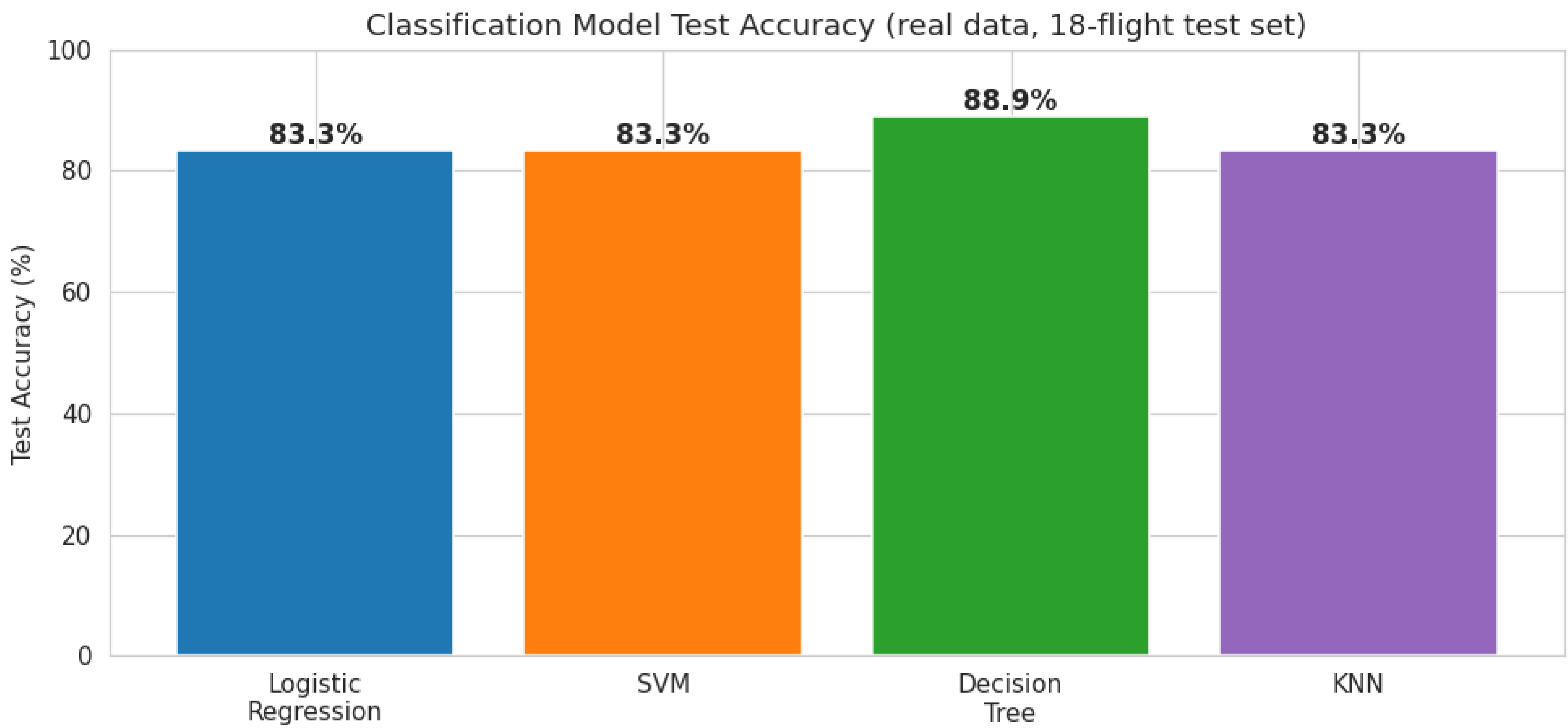
Interactive version: spacex_map.html in the repository — click a marker for exact per-site stats.

Plotly Dashboard — Results

CCAFS contributes the most successful landings by volume, but KSC and VAFB have the highest success ratios; the yearly line shows the post-2016 climb.

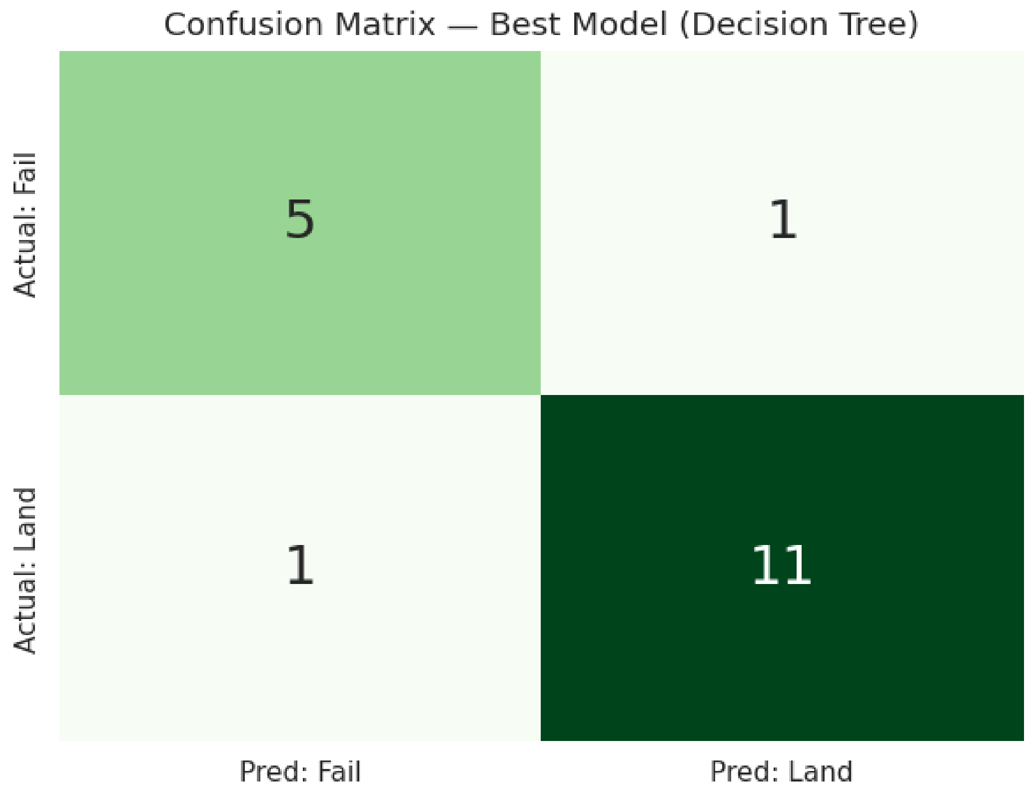
Interactive version: spacex_dash_dashboard.html in the repository.

Predictive Analysis — Model Comparison



All four models exceed 83% test accuracy. The Decision Tree is best at 88.89%; Logistic Regression, SVM and KNN each score 83.33%. Cross-validation scores cluster tightly around 84–85%, so the models agree on the signal — the small 18-flight test set separates them.

Predictive Analysis — Best Model Confusion Matrix



True Positives: 11
landings correctly predicted

True Negatives: 5
failures correctly predicted

False Positives: 1
failure predicted as a landing

False Negatives: 1
landing predicted as a failure

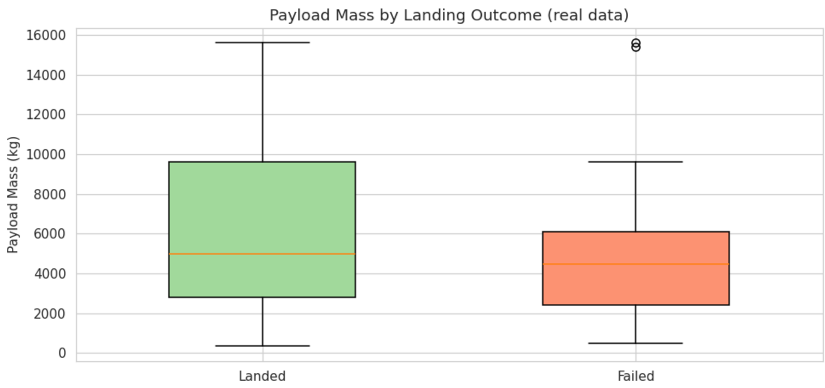
Interpretation:

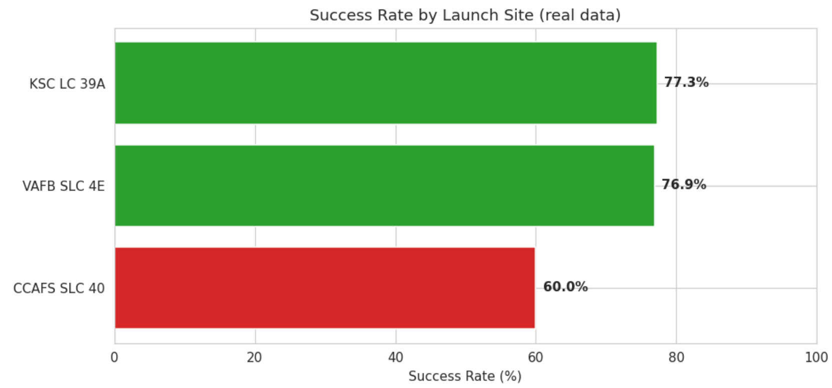
The Decision Tree correctly classifies 16 of 18 test flights (88.9%). Its main error is over-predicting landings (false positives) — the same pattern seen across all four models on this dataset.

Conclusion

- **Goal met:** a full pipeline (collection → wrangling → EDA → SQL → Folium → dashboard → classification) predicts Falcon 9 landing success at 88.89% on real data.
- **Best model:** the Decision Tree (88.89% test accuracy), narrowly ahead of Logistic Regression, SVM and KNN (all 83.33%).
- **What drives success:** accumulated experience (FlightNumber) and booster reuse matter most; orbit matters (GTO hardest at 52%, VLEO/LEO strong); launch site matters (KSC/VAFB > CCAFS).
- **Trend:** landing reliability rose from ~0% (2013) to ~90% (2019), so recent performance far exceeds the 67% all-time average.
- **Business value:** a competitor can predict landing (and therefore reuse and cost) from payload, orbit, site and booster history before a launch is booked.

Creativity & Innovative Insights





- **Counter-intuitive insight:** successful landings carry heavier payloads on average (6,765 kg vs 4,784 kg). The driver is the reused-booster Starlink era — flight-proven hardware lands heavy VLEO payloads reliably, so experience beats payload weight.
- **Learning-curve insight:** FlightNumber is the strongest single signal — early flights fail, post-2016 flights mostly land — which is why every classifier leans on it.
- **Repository:** https://github.com/Eyad-Alarfaj/spacex-capstone (all executed notebooks, real dataset, map, dashboard, and every figure in this deck).